# HAna Exam 2025/26 - An Analyst Never Forgets

## Reals, Sequences, and Series

### Real Sequences

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**Definition: Bounds and Boundedness**

Let $A \subseteq R$.
* $x \in \mathbb{R}$ is an **upper bound for** $A$ if $x \geq a$ for all $a \in A$
* $s \in \mathbb{R}$ is a **least upper bound** or **supremum** for $A$ if $s$ is an upper bound for $A$, and $x \geq s$ whenever $x$ is an upper bound for $A$.
* $A$ is **bounded** if it is bounded above and below

**Axiom: The Completeness Axiom**

Every bounded above non-empty $A \subseteq \mathbb{R}$ has a least upper bound in $\mathbb{R}$

**Definition: Nested Sets**

A sequence $(I_n)_{n\in\mathbb{N}}$ of sets is **nested** if $I_n \supseteq I_{n+1}$ for all $n \in \mathbb{N}$

**Theorem: The Nested Interval Theorem**

If $(I_n)_{n\in\mathbb{N}}$ is a nested sequence of nonempty closed bounded intervals, then the set $E := \bigcap_{n\in\mathbb{N}} I_n = \{x \in \mathbb{R} : x \in I_n \forall n \in \mathbb{N}\}$ is nonempty. If $\lambda(I_n) \to 0$ as $n \to \infty$ then $E$ contains exactly one element.

**Definition: Sequence Convergence**

A real sequence $(a_n)_{n\in\mathbb{N}}$ **converges** to $a \in \mathbb{R}$ iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$.
A sequence which does not converge to some real number **diverges**.
• We say $a_n \to \infty$ as $n \to \infty$ if $\forall R \in \mathbb{R}$, $\exists N \in \mathbb{N}$ s.t. $a_n > R$ for all $n \geq N$
• We say $a_n \to -\infty$ as $n \to \infty$ if $\forall R \in \mathbb{R}$, $\exists N \in \mathbb{N}$ s.t. $a_n < R$ for all $n \geq N$

**Major Definition: Cauchy Sequences**

A real sequence $(a_n)_{n\in\mathbb{N}}$ is **Cauchy** if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|a_n - a_m| < \epsilon$ for all $n, m \geq N$.

**Major Theorem: The Cauchy Criterion**

A real sequence is convergent if and only if it is Cauchy.

*Proof.* 1: Let $(a_n)_{n\in\mathbb{N}}$ be convergent. Then $\exists a \in \mathbb{R}$ s.t. $\lim_{n\to\infty} a_n = a$, and for all $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $|a_n - a| < \epsilon/2$ when $n \geq N$. Then by the TI, $|a_n - a_m| = |(a_n - a) + (a - a_m)| \leq |a_n - a| + |a - a_m| = |a_n - a| + |a_m - a| < \epsilon/2 + \epsilon/2 = \epsilon$ for $m, n \geq N$, thus the sequence is Cauchy.
2: Let $(a_n)_{n\in\mathbb{N}}$ be Cauchy. First show it is bdd. Let $\epsilon = 1$. Then $\exists N \in \mathbb{N}$ s.t. $|a_n - a_m| < 1$ for $m, n \geq N$. In particular $|a_m| < 1 + |a_N|$ for all $m \geq N$, so $|a_m| \leq \max\{|a_1|, \dots, |a_{N-1}|, |a_N| + 1\}$ for $m \in \mathbb{N}$, so $(a_n)_{n\in\mathbb{N}}$ is bdd. Secondly, by the B-WT, $\exists (a_{n_k})_{k\in\mathbb{N}}$ s.t. $a_{n_k} \to a \in \mathbb{R}$ as $k \to \infty$. Finally, show $a_n \to a$: Let $\epsilon > 0$. Since $(a_n)_{n\in\mathbb{N}}$ is Cauchy, $\exists N, K \in \mathbb{N}$ s.t. $|a_n - a_m| < \epsilon/2$ for $m, n \geq N$ and $\left|a_{n_k} - a\right| < \epsilon/2$ for $k \geq K$. Fix $k \in \mathbb{N}$ s.t. $k \geq K$ and $n_k \geq N$. Then $|a_n - a| \leq \left|a_n - a_{n_k}\right| + \left|a_{n_k} - a\right| < \epsilon/2 + \epsilon/2 = \epsilon$, $\forall n \geq N$. Thus $a_n \to a$ as $n \to \infty$ as required. □

**Definition: Subsequences**

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence. A **subsequence** of $(a_n)_{n\in\mathbb{N}}$ is a real sequence of the form $(a_{n_k})_{k\in\mathbb{N}}$ where for each $k \in \mathbb{N}$ there exists $n_k \in \mathbb{N}$ such that $n_1 < n_2 < \dots < n_k < n_{k+1} < \dots$.

**Theorem: Bolzano-Weierstrass Theorem**

Every bounded sequence of real numbers has a convergent subsequence.

*Proof.* Let $(a_n)_{n\in\mathbb{N}}$ be a bdd seq., with upper/lower bound $a, b \in \mathbb{R}$.
Let $I_1 = [a, b]$ and divide $I_1$ into $I_1' = [a, (a+b)/2], I_1'' = [(a+b)/2, b]$.
Since $I_1 = I_1' \cup I_1''$, at least one of $I_1', I_1''$ contains infinitely many $a_n$. Choose one with this property and denote it $I_2$. Let $n_2 \in \mathbb{N}$ be s.t. $n_2 > n_1 = 1$ and $a_{n_2} \in I_2$. Repeat inductively. By the above process we recursively define a nested sequence of intervals $(I_k)_{k\in\mathbb{N}}$ and subsequence $(a_{n_k})_{k\in\mathbb{N}}$ s.t. each $I_k$ contains infinitely many terms of $(a_n)_{n\in\mathbb{N}}$, $a_{n_k} \in I_k$, and $\lambda(I_k) = 2^{-(k-1)}(b-a)$.
Suppose in this way we construct $a_{n_1}, \dots, a_{n_k}, k \in \mathbb{N}$, and intervals $I_1, \dots, I_k$. Divide $I_k$ into $I_k', I_k''$ as above, again at least one contains infinitely many $a_n$. Choose one with this property, denote it $I_{k+1}$. Let $n_{k+1} \in \mathbb{N}$ be s.t. $n_{k+1} > n_k$, $a_{N_{k+1}} \in I_{k+1}$.
Now observe $(I_k)_{k\in\mathbb{N}}$ is a nested seq. of clounded intervals, so by the NIT there is $x \in \mathbb{R}$ s.t. $x$ lies in all $I_k$. Since $a_{n_k} \in I_k$, $\left|x - a_{n_k}\right| \leq \lambda(I_k) = \frac{b-a}{2^{k-1}}$ for all $k \in \mathbb{N}$. Then by the Squeeze Theorem, $a_{n_k} \to x$ as $k \to \infty$. □

### Series

**Major Definition: Infinite Series**

Let $(a_k)_{k\in N}$ be a real sequence. We form the **sequence of partial sums** $(s_n)_{n\in\mathbb{N}}$ by $s_n := \sum_{k=1}^{n} a_k$ for all $n \in \mathbb{N}$.
• $\sum_{k=1}^{\infty} a_k$ **converges to** $a$ if $\lim_{n\to\infty} s_n$ exists and $= a$, else **diverges**.

**Definition: Absolute Convergence**

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence. Then $\sum_{n=1}^{\infty} a_n$:
(1) **Converges absolutely** if $\sum_{n=1}^{\infty} |a_n|$ converges
(2) **Converges conditionally** if it converges but $\sum_{n=1}^{\infty} |a_n|$ diverges.

**Theorem: Absolute Convergence vs Convergence**

Absolute convergence implies convergence.
Note the converse does not always hold (consider $\sum_{n=0}^{\infty} (-1)^n/n$.

*Proof.* Let $(a_k)_{k=1}^{\infty}$ be a real seq., and $\sum_{n=1}^{\infty} a_n$ be absolutely convergent. We show the series satisfies the Cauchy criterion: Let $\epsilon > 0$. Since $\sum_{n=1}^{\infty}$ cnvges absolutely, by the Cauchy criterion for series $\exists N \in \mathbb{N}$ s.t. $\sum_{k=m+1}^{n} |a_k| < \epsilon$ for all $n > m \geq N$. Now let $n > m \geq N$ and apply the TI repeatedly to obtain $\left|\sum_{k=m+1}^{n} a_k\right| \leq \sum_{k=m+1}^{n} |a_k| < \epsilon$. This establishes the Cauchy criterion for $\sum_{n=1}^{\infty} a_n$, so the series is convergent. □

**Major Theorem: The Cauchy Criterion For Series**

Let $(a_k)_{k=1}^{\infty}$ be a real sequence. Then $\sum_{k=1}^{\infty} a_k$ converges iff for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\left|\sum_{k=m+1}^{n} a_k\right| < \epsilon$ for all $n > m \geq N$.

*Proof.* Let $(s_n)_{n\in\mathbb{N}}$ be the seq. of partial sums of the series. Observe that $s_n - s_m = \sum_{k=1}^{n} a_k - \sum_{k=1}^{m} a_k = \sum_{k=m+1}^{n} a_k$ for $n > m$. Then the series satisfies the Cauchy criterion iff $(s_n)_{n\in\mathbb{N}}$ is Cauchy. $(s_n)_{n\in\mathbb{N}}$ is Cauchy iff it is convergent, and convergent iff $\sum_{k=1}^{\infty} a_k$ converges. □

**Theorem: The Riemann Rearrangement Theorem**

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence and suppose $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. Then for *any* $s \in \mathbb{R}$, there exists a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that the rearrangement $\sum_{n=1}^{\infty} a_{\sigma(n)}$ converges to $s$.

Note the RRT does not apply to absolutely convergent series. Indeed, for $\sum_{n=1}^{\infty} a_n$ absolutely convergent, and any bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is convergent and $\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$

### Subsets Of The Real Line

**Definition: Open Sets**

$U \subseteq \mathbb{R}$ is **open** if for all $x \in U$ there exists $r > 0$ s.t. $(x-r, x+r) \subseteq U$.
• The union of any collection of open sets is open
• The intersection of any finite collection of open sets is open
• Any nonempty open $U \subseteq \mathbb{R}$ is a finite or countable union of nonempty disjoint open intervals.
• If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) > \lambda\}$ is open.
• For $U \subseteq \mathbb{R}$, $f : U \to \mathbb{R}$ cont., $\lambda \in \mathbb{R}$, $\{x \in U : f(x) > \lambda\}$ is open.

**Definition: Closed Sets**

A set $F \subseteq \mathbb{R}$ is **closed** if its complement $\mathbb{R} \setminus F$ is open.
• The union of any finite collection of closed sets is closed
• The intersection of any collection of closed sets is closed
• If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) \leq \lambda\}$ is closed.

**Definition: Sequential Closure**

$F \subseteq \mathbb{R}$ is **sequentially closed** if for every convergent sequence $(x_n)_{n\in\mathbb{N}}$ s.t. $x_n \in F$ for all $n \in \mathbb{N}$, the limit $x := \lim_{n\to\infty} x_n$ satisfies $x \in F$.
• $F \subseteq \mathbb{R}$ is closed if and only if it is sequentially closed.

*Proof.* First prove by contrapositive closed sets are sequentially closed: Let $F$ be closed and $(x_n)_{n\in\mathbb{N}}$ convergent s.t. $x := \lim_{n\to\infty} x_n \notin F$. We show $\exists N \in \mathbb{N}$ s.t. $x_N \notin F$. $F$ is closed, so by definition $U := \mathbb{R} \setminus F$ is open. Moreover $x \in U$, so $\exists \epsilon > 0$ s.t. $(x-\epsilon, x+\epsilon \subseteq U)$, and $N \in \mathbb{N}$ s.t. $|x_n - x| < \epsilon \forall n \in \mathbb{N}$. In particular $x_N \in (x-\epsilon, x+\epsilon) \subseteq U$, so $x_N \notin F$.
Now prove by contradiction sequentially closed sets are closed: Suppose $F$ is not closed, equivalently $U := \mathbb{R} \setminus F$ is not open. Then $\exists x \in U$ s.t. $\forall r > 0$, $(x-r, x+r) \cap F \neq \emptyset$. For $n \in \mathbb{N}$, let $x_n \in ((x - \frac{1}{n}, x + \frac{1}{n}) \cap F)$. Then $(x_n)_{n\in\mathbb{N}}$ converges to $x$, since $|x - x_n| \leq \frac{1}{n}$. In particular, since $F$ is sequentially closed, we must have $x \in F$, a contradiction. □

**Definition: The Middle Thirds Cantor Set**

The **Middle Thirds Cantor Set** $C$ is a closed set defined as follows:
(1) Divide [0,1] into thirds, remove the open middle third interval $(\frac{1}{3}, \frac{2}{3})$.
(2) Divide the remaining two intervals into thirds and remove the open middle thirds intervals $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$.
(3) Repeat this process ad infinitum.
At the $n^{th}$ stage of the construction we have $2^n$ closed intervals $(I_{n,k})_{k=1}^{2n}$ with $\lambda(I_{n,k}) = 3^{-n}$ for all $1 \leq k \leq 2^n$. Then $C := \bigcap_{n=1}^{\infty} C_n$ where $C_n := +\bigcup_{k=1}^{2^n} I_{n,k}$ for $n \in \mathbb{N}$. Each $I_{n,k}$ is closed, so their finite union $C_n$ and the intersection $C$ of these $C_n$ are also closed.

**Example: Showing Not Closed And Not Open**

Show $S := \{e^k : k \in \mathbb{Z}\}$ is neither open nor closed.
First show $S$ is not open:
Let $k = 0$. $e^0 = 1$, so $1 \in S$. For $k > 0$, $e^k \geq e^1$, and for $k < 0$, $e^k \leq e^{-1}$. Thus $S \cap (e^{-1}, e) = \{1\}$. Since $\{1\}$ is not open, $S$ cannot be open, as $(e^{-1}, e)$ is open and a finite intersection of open sets is open.
Now show $S$ is not closed: The seq $x_n := e^{-n}$ has $x_n \in S$ for all $n \in \mathbb{N}$ but $\lim_{n\to\infty} x_n = 0 \notin S$. Thus $S$ is not sequentially closed, ie not closed.

**Ex.: A Seq With Bounded Term Differences Is Cauchy**

Let $(a_n)_{n\in\mathbb{N}}$ be s.t. $\left|a_n - a_{n-1}\right| \leq 2^{-n}$. Show $(a_n)_{n\in\mathbb{N}}$ is Cauchy.
Consider $|a_n - a_m|$. Form the telescoping sum then apply the TI: $|a_n - a_m| \leq \left|a_n - a_{n-1}\right| + \left|a_{n-1} - a_{n-2}\right| + \dots + \left|a_{m+1} - a_m\right| \leq \frac{1}{2^n} + \frac{1}{2^{n-1}} + \dots + \frac{1}{2^{m+1}}$. Thus $|a_n - a_m| \leq \sum_{k=m+1}^{n} \frac{1}{2^k} \leq \frac{1}{2^m}$, so given $\epsilon > 0$ choose $N \in \mathbb{N}$ s.t. $\frac{1}{2^N} < \epsilon$ and the seq is Cauchy.

## Measure And Integration

### The Lebesgue Measure

**Definition: Length**

For a nonempty bounded interval $I \subseteq \mathbb{R}$, we define the **length** $\lambda$ **of I** as $\lambda(I) = \sup(I) - \inf(I)$. $\lambda(\emptyset) = 0$ and for $I$ unbounded, $\lambda(I) = \infty$.

**Major Definition: Lebesgue Measurable Sets**

$E \subseteq \mathbb{R}$ is **Lebesgue measurable** if for all $\epsilon > 0$, there exists open $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m(U \setminus E) < \epsilon$. The collection of all L-mesable subsets of $\mathbb{R}$ is $\mathcal{M} \subseteq \mathcal{P}(\mathbb{R})$. We only handle L-mesable sets in this course thankfully!

**Major Definition: The Lebesgue Measure**

The **Lebesgue measure** is the function $m : \mathcal{M} \to [0, \infty]$. For $E \in \mathcal{M}$, $m(E) := \inf\left\{\sum_{j=1}^{\infty} \lambda(I_j) : E \subseteq \bigcup_{j=1}^{\infty} I_j, I_j \subseteq \mathbb{R} \text{ an open interval}\right\}$. Here the inf is over all countable coverings $(I_j)_{j\in\mathbb{N}}$ of $E$ by open intervals.
* $m(\emptyset) = 0$
* If $E \subseteq F \subseteq \mathbb{R}$, $m(E) \leq m(F)$
* If $(E_j)_{j\in\mathbb{N}}$ is any seq of sets in $\mathcal{M}$, $m\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} m(E_j)$.
* If the $E_j$ above are all disjoint then $m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j)$
• If $\sum_{j=1}^{\infty} \lambda(I_j)$ diverges, we say it has value $\infty$. If there is no cover of $E$ s.t. $\sum_{j=1}^{\infty} \lambda(I_j) < \infty$, $m(E) = \infty$.
• For any interval $J \subseteq \mathbb{R}$, $\lambda(J) = m(J)$.
• Given $E \subseteq \mathbb{R}$, $\epsilon > 0$, $\exists$ open $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m(U) < m(E) + \epsilon$.

**Definition: The Lebesgue Outer Measure**

The **Lebesgue Outer Measure** is $m^* : \mathcal{P} \to [0, \infty]$. We base our definition of $m$ on this, $m^*(E) = m(E)$ for $E \in \mathcal{M}$. For $E \notin \mathcal{M}$ it is only countably subadditive (ie measure of union $\leq$ sum of measures).

**Definition: Null Sets**

A set $E \subseteq \mathbb{R}$ is **null** or **has measure zero** if $m(E) = 0$.
Every countable subset of $R$ has measure zero.

*Proof.* First prove any singleton set $\{x\}$ for $x \in \mathbb{R}$ has measure zero. Given $x \in \mathbb{R}$, let $\epsilon > 0$ and consider the cover $I := (x - \epsilon/2, x + \epsilon/2)$. Then $m(\{x\}) \leq \lambda(I) = \epsilon$. Since $\epsilon > 0$ is arbitrary we conclude $m(\{x\}) = 0$. Any countable $E$ is a countable union of singletons, $E = \bigcup_{j=1}^{\infty} \{x_j\}$.
Apply the countable subadditivity property to conclude $m(E) = 0$. □

**Definition: Distance Between Sets**

For $E, F \subseteq \mathbb{R} \neq \emptyset$ and disjoint; $\text{dist}(E, F) := \inf\{|x - y| : x \in E, y \in F\}$.
• If $E, F \subseteq \mathbb{R} \neq \emptyset$, disjoint, $\text{dist}(E, F) > 0$; $m*(E \cup F) = m(E) + m(F)$
• If $F, K \subseteq \mathbb{R}$ nonempty, closed, $F \cap K = \emptyset$, $K$ bounded; then $\text{dist}(F, K) > 0$.
• If $(K_j)_{j\in\mathbb{N}}$ a seq of disjoint closed $K_i$; $m\left(\bigcup_{j=1}^{\infty} K_j\right) = \sum_{j=1}^{\infty} m(K_j)$

**Theorem: Properties Of L-Measurable Sets**

• Any open set is L-measurable
• Let $F \subseteq \mathbb{R}$ and $(E_j)_{j\in\mathbb{N}}$ be L-mesable. Then:
(1) The union $E := \bigcup_{j=1}^{\infty} E_j$ is L-mesable
(2) The complement $\mathbb{R} \setminus F$ is L-mesable
(3) The intersection $\bigcap_{j=1}^{\infty} E_j$ is L-mesable
• Let $E, F \subseteq \mathbb{R}$ be L-mesable. Then $F \setminus E$ is L-mesable
• For $E \subseteq \mathbb{R}$, $E$ is L-mesable iff $\forall \epsilon > 0$, $\exists$ closed $K \subseteq E$ s.t. $m(E \setminus K) < \epsilon$.

**Theorem: Monotone Convergence For Measure**

Let $(E_j)_{j\in\mathbb{N}}$ be a seq. of L-mesable sets s.t. $E_j \subseteq E_{j+1}$ $\forall j \in \mathbb{N}$.
Then $m\left(\bigcup_{j=1}^{\infty} E_j\right) = \lim_{j\to\infty} m(E_j)$

### The Lebesgue Integral

**Definition: Characteristic Functions Of Sets**

For $E \subseteq \mathbb{R}$, the **characteristic function of** $E$ is $\chi_E : \mathbb{R} \to \mathbb{R}$, $\chi_E(x) := 1$ if $x \in E$; $:= 0$ if $x \notin E$.

**Definition: Simple Functions**

$s : \mathbb{R} \to \mathbb{R}$ is **simple** if it is a linear combination of characteristic functions of L-mesable sets. Ie, $\exists c_1, \dots, c_k \in \mathbb{R}$ and L-mesable $E_1, \dots, E_k \subseteq \mathbb{R}$ s.t. $s(x) = \sum_{j=1}^{k} c_j \chi_{E_j}(x)$ for all $x \in \mathbb{R}$.

**Definition: L-Integral Of A Non-Negative Simple Function**

Let $s : \mathbb{R} \to [0, \infty)$ be simple, $a_j \in [0, \infty), E_j \in \mathcal{M}$, $s = \sum_{j=1}^{k} a_j \chi_{E_j}$.
The **Lebesgue integral of** $s$ is $\int_{\mathbb{R}} s(x)\, dx := \sum_{j=1}^{k} a_j m(E_j) \in [0, \infty)$.
• $\int_{\mathbb{R}} \chi_E(x)\, dx := m(E)$
• For $E_j$ with infinite measure, the integral takes value $\infty$. If $a_j = 0$ and $\lambda(E_j) = \infty$, use the convention $0 \cdot \infty = 0$.

**Major Definition: Lebesgue Measurable Functions**

Let $E \subseteq \mathbb{R}$ be L-mesable. $f : \mathbb{E} \to \mathbb{R}$ is **Lebesgue measurable** if for all $\lambda \in \mathbb{E}$, the strict superlevel set $\{x \in \mathbb{E} : f(x) > \lambda\}$ is L-mesable
• Any continuous function $f : \mathbb{R} \to \mathbb{R}$ is L-mesable
• If $E \subseteq \mathbb{R}$, $f, g : \mathbb{R} \to \mathbb{R}$ are all L-mesable; then $\max\{f, g\}$, $\min\{f, g\}$, $f \cdot \chi_E$ are all L-mesable
• If $f, g : \mathbb{R} \to \mathbb{R}$ L-mesable, $\alpha, \beta \in \mathbb{R}$; then $\alpha f + \beta g$, $f \cdot g$ are L-mesable.

**Theorem: Approxing Non-Negative L-Measurable Functions**

Let $f : \mathbb{R} \to [0, \infty)$ be a non-negative L-mesable func. Then there exists a sequence $s_n : \mathbb{R} \to [0, \infty)$ of non-negative simple functions s.t.:
(1) $(s_n)_{n\in\mathbb{N}}$ is ptwise non-decreasing - $0 \leq s_n \leq s_{n+1}$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$.
(2) $s_n \to f$ pointwise on $\mathbb{R}$ as $n \to \infty$.

**Dfn: Integrating Non-Negative L-Measurable Functions**

Let $f : \mathbb{R} \to [0, \infty)$ be a non-negative L-mesable func. The **Lebesgue integral of** $f$ as $\int_{\mathbb{R}} f(x)\, dx := \sup\left\{\int_{\mathbb{R}} s(x)\, dx : s \text{ simple}, 0 \leq s \leq f\right\}$ where the supremum is taken over all non-negative simple functions $s : \mathbb{R} \to \mathbb{R}$ with $s(x) \leq f(x)$ for all $x \in \mathbb{R}$.
• If $f : \mathbb{R} \to [0, \infty)$, $E \subseteq \mathbb{R}$ both L-mesable, $\int_E f(x)\, dx := \int_{\mathbb{R}} f(x) \chi_E(x)\, dx$
• For L-mesable $f$ defined only on L-mesable $E \subseteq \mathbb{R}$, define $\int_E f(x)\, dx = \int_{\mathbb{R}} \tilde{f}(x)\, dx$, with $\tilde{f} : \mathbb{R} \to [0, \infty)$, $\tilde{f}(x) := f(x)$ if $x \in E$, 0 if $x \in \mathbb{R} \setminus E$.
• Let $f, g : \mathbb{R} \to [0, \infty)$ be L-mesable. Then:
(1) If $0 \leq f(x) \leq g(x)$ for all $x \in \mathbb{R}$, $0 \leq \int_{\mathbb{R}} f(x)\, dx \leq \int_{\mathbb{R}} g(x)\, dx$
(2) For all $\lambda > 0$, $m(\{x \in \mathbb{R} : f(x) > \lambda\}) \leq \frac{1}{\lambda} \int_{\mathbb{R}} f(x)\, dx$ (Markov's Ineq.)
• This L-integral respects linearity
• If $f : \mathbb{R} \to [0, \infty)$ is L-mesable and $E, F \subseteq \mathbb{R}$ are disjoint L-mesable sets, $\int_{E \cup F} f(x)\, dx = \int_E f(x)\, dx + \int_F f(x)\, dx$
• Let $h_k : \mathbb{R} \to [0, \infty)$ be a seq of non-neg L-mesable funcs on $\mathbb{R}$ s.t. $\sum_{k=1}^{\infty} h_k$ cnvges ptwise. Then $\sum_{k=1}^{\infty} \int_{\mathbb{R}} h_k(x)\, dx = \int_{\mathbb{R}} \sum_{k=1}^{\infty} h_k(x)\, dx$.

**Theorem: Pointwise Limits And L-Integration**

The pointwise limit of L-integrals is not necessarily the L-integral of the pointwise limit. Ie, $\exists$ L-mesable $f_n, f : \mathbb{R} \to [0, \infty)$ s.t.:
• $f_n$ is L-integrable for all $n \in \mathbb{N}$ and $f$ is L-integrable
• $f_n \to f$ pointwise on $(0, 1]$
• $\int_0^1 f(x)\, dx \neq \lim_{n\to\infty} \int_0^1 f_n(x)\, dx$,
ie $\lim_{n\to\infty} \int_0^1 f_n(x)\, dx \neq \int_0^1 (\lim_{n\to\infty} f_n(x))\, dx$
• For counterexample: $f, f_n : \mathbb{R} \to [0, \infty)$, $f(x) := \mathbb{R} \to [0, \infty)$, $f_n(x) := n$ if $0 < x \leq \frac{1}{n}$ and 0 otherwise.

**Major Theorem: The MCT For Lebesgue Integrals**

Let $f_k : \mathbb{R} \to [0, \infty)$ be a ptwise non-decreasing seq. of non-negative L-mesable funcs converging ptwise to $f := \lim_{k\to\infty} f_k : \mathbb{R} \to [0, \infty)$. Then $f$ is L-mesable, and $\lim_{k\to\infty} \int_{\mathbb{R}} f_k(x)\, dx = \int_{\mathbb{R}} \lim_{k\to\infty} f_k(x)\, dx = \int_{\mathbb{R}} f(x)\, dx$. Note both sides may equal $\infty$.
• For $f, g : \mathbb{R} \to [0, \infty)$ L-mesable and $\alpha, \beta \geq 0$, the integral respects linearity
• If $f : \mathbb{R} \to [0, \infty)$ L-mesable and $E, F \subseteq \mathbb{R}$ disjoint, $\int_{E \cup F} f(x)\, dx = \int_E f(x)\, dx + \int_F f(x)\, dx$.
• For $h_k : \mathbb{R} \to [0, \infty)$ a seq. of L-mesable funcs s.t. $\sum_{k=1}^{\infty} h_k$ converges pointwise, $\sum_{k=1}^{\infty} \int_{\mathbb{R}} h_k(x)\, dx = \int_{\mathbb{R}} \sum_{k=1}^{\infty} h_k(x)\, dx$.

**Lemma: The Fatou Lemma**

If $f_k : \mathbb{R} \to [0, \infty)$ is a seq of non-neg L-mesable funcs, then $\int_{\mathbb{R}} \text{liminf}_{k\to\infty} f_k(x)\, dx \leq \text{liminf}_{k\to\infty} \int_{\mathbb{R}} f_k(x)\, dx$.

**Lemma: The Reverse Fatou Lemma**

Let $f_k : \mathbb{R} \to [0, \infty)$ be a seq of non-neg L-mesable funcs and $g : \mathbb{R} \to [0, \infty)$ L-intable s.t. $f_k(x) \leq g(x)$ for all $x \in \mathbb{R}$. Then $\int_{\mathbb{R}} \text{limsup}_{k\to\infty} f_k(x)\, dx \geq \text{limsup}_{k\to\infty} \int_{\mathbb{R}} f_k(x)\, dx$.

**Dfn: Decomposing L-Measurable Real-Valued Functions**

Let $f : \mathbb{R} \to \mathbb{R}$ be L-mesable. $f^+(x) := \max\{f(x), 0\}$ and $f^-(x) := -\min\{f(x), 0\}$, $f = f^+ - f^-$, and $|f| = f^+ + f^-$ are all L-mesable.
Moreover, $\int_{\mathbb{R}} |f(x)|\, dx < \infty \iff \int_{\mathbb{R}} f^+(x)\, dx < \infty$ and $\int_{\mathbb{R}} f^-(x)\, dx < \infty$.

**MDfn: Lebesgue Integration Of Real-Valued Functions**

An L-mesable $f : \mathbb{R} \to \mathbb{R}$ is **Lebesgue integrable** if $\int_{\mathbb{R}} |f(x)|\, dx < \infty$.
The collection of all L-integrable functions is $\mathcal{L}^1(\mathbb{R})$.
For L-integrable $f : \mathbb{R} \to \mathbb{R}$, $\int_{\mathbb{R}} f(x)\, dx := \int_{\mathbb{R}} f^+(x)\, dx - \int_{\mathbb{R}} f^-(x)\, dx$.
• For $f \in \mathcal{L}^1(\mathbb{R})$, $|\int_{\mathbb{R}} f(x)\, dx| \leq \int_{\mathbb{R}} |f(x)| dx$
• This L-integral respects linearity for $\alpha, \beta \in \mathbb{R}$.
• For L-mesable $E \subseteq \mathbb{R}$, $\mathcal{L}^1(E) = \left\{f : E \to \mathbb{R} : f \text{ is L-mesable}, \tilde{f} \in \mathcal{L}^1(\mathbb{R})\right\}$.

**Major Theorem: The Dominated Convergence Theorem**

Let $f_k : \mathbb{R} \to \mathbb{R}$ be a seq. of L-integrable funcs converging ptwise to $f : \mathbb{R} \to \mathbb{R}$, and $g \in \mathcal{L}^1(\mathbb{R})$ be s.t. $|f_k(x)| \leq g(x)$ for all $x \in \mathbb{R}, k \in \mathbb{N}$. Then $f \in \mathcal{L}^1(\mathbb{R})$ and $\lim_{k\to\infty} \int_{\mathbb{R}} f_k(x)\, dx = \int_{\mathbb{R}} \lim_{k\to\infty} f_k(x)\, dx = \int_{\mathbb{R}} f(x)\, dx$.
• Uniform Convergence Theorem: Let $E \subseteq \mathbb{R}$ be L-mesable s.t. $m(E) < \infty$, and $f_n, f : \mathbb{R} \to \mathbb{R}$ L-integrable funcs s.t. $f_n \to f$ unifly on $E$ as $n \to \infty$, Then $f \in \mathcal{L}^1(\mathbb{R})$ and $\lim_{n\to\infty} \int_E f_n(x)\, dx = \int_E \lim_{n\to\infty} f_n(x)\, dx$

**Theorem: The Fundamental Theorem Of Calculus**

Let $a, b \in \mathbb{R}$ with $a < b$ and $f : [a, b] \to \mathbb{R}$ continuous. Then $F(x) := \int_a^x f(y)\, dy$ is diffable on $[a, b]$ and $F'(x) = f(x)$ for all $x \in [a, b]$
• For $a, b \in \mathbb{R}$, $a < b$, $f : [a, b] \to \mathbb{R}$ cont.sly diffable; $\int_a^b f'(x)\, dx = f(b) - f(a)$

# Uniform Continuity, Uniform Convergence, Power Series

## Uniform Continuity

### Definition: Continuity

For $I \subseteq \mathbb{R}$ an interval and $f: I \to \mathbb{R}$, $f$ is **continuous** if for all $x_0 \in I$, $\epsilon > 0$, there exists $\delta > 0$ s.t. $|f(x) - f(x_0)| < \epsilon \forall x$ s.t. $|x - x_0| < \delta$.
- A cont func. on $[a,b] \subseteq \mathbb{R}$ is bounded and attains a max/min on $[a,b]$.

### Major Definition: Uniform Continuity

Let $I \subseteq \mathbb{R}$ be an interval and $f: I \to \mathbb{R}$. $f$ is **uniformly continuous** if $\forall \epsilon > 0, \exists \delta > 0$ s.t. $|f(x) - f(y)| < \epsilon \ \forall x, y \in I$ where $|x - y| < \delta$. Note the key difference with regular continuity - here $\delta$ may only depend on $\epsilon$, not $x$.

### Lemma: Helpful Uniform Continuity Stuff

- If you can bound the deriv of a diffable func, by MVT it is uniformly continuous (need to prove, for deriv bdd by $M$, use $\delta = \epsilon/M$.).
- For all $x \in \mathbb{R}$, $-\frac{\pi}{2} < \arctan(x) < \frac{\pi}{2}$ and $1 \geq \arctan'(x) = \frac{1}{x^2+1} > 0$

### Theorem: The Heine-Cantor Theorem

If $a < b$ and $f: [a,b] \to \mathbb{R}$ is continuous, then $f$ is uniformly continuous.

### Example: Proving Uniform Continuity

Show $f: (0,\infty) \to \mathbb{R}, f(x) := \frac{\sin(x)}{x}$ is uniformly continuous.
$f'(x) = \frac{x\cos(x) - \sin(x)}{x^2} = \frac{x(\cos(x)-1) + (x - \sin(x))}{x^2}$ for $x > 0$.
WLOG assume $x < y$. Apply MVT to $x \mapsto \cos(x)$. For $x > 0 \exists c \in (0,x)$ s.t. $\cos(x) - 1 = -\sin(c)x$, so $|\cos(x) - 1| = |-\sin(c)x| \leq |x|$.
Apply MVT to $x \mapsto x - \sin(x)$. For $x > 0, \exists d \in (0,x)$ s.t. $x - \sin(x) = (1 - \cos(d))x$. As above $|1 - \cos(d)| \leq |x|$, so $|x - \sin(x)| \leq |x||x| = x^2$.
Thus for all $x > 0$, $\left|f'(x)\right| = \frac{|x(\cos(x)-1)| + |x - \sin(x)|}{x^2} \leq \frac{x^2 + x^2}{x^2} = 2$
Then for all $x, y \in (0,\infty) \exists \xi \in (x,y)$ s.t. $|f(x) - f(y)| = \left|f'(\xi)\right||x - y| \leq 2|x - y|$ so let $\delta = \epsilon/2$, then $|f(x) - f(y)| \leq 2|x - y| < 2\delta = \epsilon$.

### Example: Disproving Uniform Continuity

Show $f: (0,\infty) \to \mathbb{R}, f(x) = 1/x^2$ is not uniformly continuous.
It suffices to show for all $\delta > 0$ there exist $x, y \in \mathbb{R}$ s.t. $|x - y| < \delta$ and $|f(x) - f(y)| \geq 1$. Let $0 < x < y \leq 0$. Observe $x + y > x$ and $x^2 y^2 \leq x^2$. Then $\frac{1}{x^2} - \frac{1}{y^2} = \frac{y^2 - x^2}{x^2 y^2} = \frac{(x-y)(x+y)}{x^2 y^2} \geq \frac{y - x}{x} > 0$.
Then given $\delta > 0$ let $x = \min\{\delta, 1\}/2$ and $y = 2x$.
Then $|x - y| = \min\{\delta, 1\}/2 < \delta$ and $\left|\frac{1}{x^2} - \frac{1}{y^2}\right| \geq \frac{y - x}{x} = \frac{x}{x} = 1$.
Key points - do inequalities in reverse to proving; bound $|f(x) - f(y)|$ below; $x, y$ may depend on $\delta$; can use any $x, y$ in the domain of $f$.

### Ex.: Diffable On Open Intvl, Bdd Deriv $\implies$ Unif Cont

Let $I \subseteq \mathbb{R}$ be an open interval and $f: I \to \mathbb{R}$ be differentiable with bounded derivative on $I$. Show $f$ is uniformly continuous on $I$.
$\exists M \in \mathbb{R}, \left|f'(x)\right| \leq M$ for all $x \in I$. By the MVT, for all $x, y \in \mathbb{R}$, there exists $c$ between $x$ and $y$ s.t. $|f(x) - f(y)| = \left|f'(c)\right||x - y| \leq M|x - y|$.
Then given $\epsilon > 0$, let $\delta = \epsilon/M$, and we have unif cont.ty by the definition.

## Uniform Convergence

### Definition: Pointwise Convergence

Let $E \subseteq \mathbb{R}$ be nonempty. A sequence of functions $f_n: E \to \mathbb{R}$ **converges pointwise on $E$ to** $f: E \to \mathbb{R}$ if for all $x \in E$ and $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$.
When proving pointwise convergence can just take lim of an expression as $n \to \infty$ to assert pointwise convergence to the lim (if it exists).

### Theorem: Pointwise Convergence Theorems

- The pointwise limit of bounded functions is not always bounded
  - (From exercise; counterxmp $f_n(x) = n/(nx+1), f(x) = 1/x$ on $(0,1)$)
- The ptwise limit of cont funcs is not always cont. $\exists f_n, f: [0,1] \to \mathbb{R}$ s.t.
  - $f_n$ is continuous for all $n \in \mathbb{N}$
  - $f_n \to f$ pointwise on $[0,1]$ as $n \to \infty$
  - $f$ is not continuous
  - Example: $f_n(x) := 0$ if $0 \leq x \leq \frac{1}{2} - \frac{1}{2n}$; $nx - \frac{n-1}{2}$ if $\frac{1}{2} - \frac{1}{2n} \leq x \leq \frac{1}{2} + \frac{1}{2n}$; 1 if $\frac{1}{2} + \frac{1}{2n} \leq x \leq 1$
- Ptwise lim of diffable funcs not always diffable. $\exists f_n, f: (-1,1) \to \mathbb{R}$ s.t.:
  - $f_n$ is differentiable for all $N \in \mathbb{N}$
  - $f_n \to f$ pointwise on $(-1,1)$ as $n \to \infty$
  - $f$ is not differentiable
  - Example: Let $f_n(x) := \sqrt{x^2 + \frac{1}{n}}$, limit is not diffable at 0.
- The pointwise limit of derivatives is not always the derivative of the pointwise limit. $\exists f_n, f: (-1,1) \to \mathbb{R}$ s.t.
  - $f_n$ is diffable for all $n \in \mathbb{N}$ and $f$ is diffable
  - $f_n \to f$ pointwise on $(-1,1)$
  - $f'(0) \neq \lim_{n\to\infty} f_n'(0)$, ie $\left(\lim_{n\to\infty} f_n\right)'(0) \neq \lim_{n\to\infty} f_n'(0)$, ie the derivate and ptwise lim operations do not commute.
  - Example: Let $f_n(x) = \frac{\sin(2\pi nx)}{2\pi n}, f_n' = \cos(2\pi nx), f = 0$ on $(-1,1)$.

### Definition: Graph and Vertical Neighbourhood

Let $f: E \to \mathbb{R}$. The **graph of** $f$ is the set $\Gamma(f) := \{(f, f(x) : x \in E)\} \subset \mathbb{R}^2$.
For $\epsilon > 0$, the **$\epsilon$-vertical neighbourhood of the graph of** $f$ is the set $\mathcal{N}(f, \epsilon) := \{(x,y) : x \in E, |y - f(x)| < \epsilon\} \subset \mathbb{R}^2$.

### Major Definition: Uniform Convergence

Let $E \subseteq \mathbb{R}$ be nonempty. A sequence of functions $f_n: E \to \mathbb{R}$ **converges uniformly on $E$ to** $f: E \to \mathbb{R}$ if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $x \in E, n \geq N$.
- Uniform convergence implies pointwise convergence.
  For counterexample of converse use $f_n, f: (0,1] \to \mathbb{R}$. $f_n(x) := 1$ if $0 < x < \frac{1}{n}$; 0 if $\frac{1}{n} \leq x \leq 1$ and $f(x) := 0$.

### Major Theorem: Uniform Convergence Equivalencies

Let $E \subseteq \mathbb{R}$ be nonempty, $f_n: E \to \mathbb{R}$ be a sequence of funcs, and $f: E \to \mathbb{R}$.
- The following are equivalent:
  (1) $f_n \to f$ uniformly on $E$ as $n \to \infty$
  (2) $\sup_{x \in E} |f_n(x) - f(x)| \to 0$ as $n \to \infty$
  (3) For all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $\Gamma(f_n) \subset \mathcal{N}(f, \epsilon)$ for all $n \geq N$.
  Let $f_n \to f$ uniformly on $E$ as $n \to \infty$. If $(x_n)_{n\in\mathbb{N}}$ is a real sequence with $x_n \in E$ for all $n \in \mathbb{N}$, then $|f_n(x_n) - f(x_n)| \to 0$ as $n \to \infty$.

### Theorem: Sequence Test For Uniform Convergence

Let $E \subseteq \mathbb{R}$ be nonempty, $f_n: E \to \mathbb{R}$ a seq. of funcs, $f: E \to \mathbb{R}$, and $f_n \to f$ uniformly on $E$ as $n \to \infty$. If $(x_n)_{n\in\mathbb{N}}$ a real seq. with $x_n \in E$ for all $n \geq \mathbb{N}$, $|f_n(x_n) - f(x_n)| \to 0$ as $n \to \infty$.

### Definition: Uniformly Cauchy

Let $E \subseteq \mathbb{R}$ be nonempty and $f_n: E \to \mathbb{R}$ be a sequence of functions. We say $(f_n)_{n\in\mathbb{N}}$ is **uniformly Cauchy on** $E$ if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f_m(x)| < \epsilon$ for all $x \in E$ and $m, n \geq N$.

### Major Theorem: The Uniform Cauchy Criterion

$(f_n)_{n\in\mathbb{N}}$ converges unifly on $E$ iff $(f_n)_{n\in\mathbb{N}}$ is unifly Cauchy on $E$.

### Thm: Uniform Limits Of Continuous/Diffable Functions

- A unif lim of diffable funcs is not always diffable. $\exists f_n, f: (-1,1) \to \mathbb{R}$ s.t.:
  - $f_n$ is differentiable for all $n \in \mathbb{N}$
  - $f_n \to f$ uniformly on $(-1,1)$ as $n \to \infty$
  - $f$ is not differentiable
  - Example: consider $f_n(x) := \sqrt{x^2 + (1/n)}$
- Let $I \subseteq \mathbb{R}$ be a bdd open interval, $f_n: I \to \mathbb{R}$ a seq of diffable funcs s.t.:
  (1) There exists $x_0 \in I$ s.t. $(f_n(x_0))_{n\in\mathbb{N}}$ converges
  (2) There exists $g: I \to \mathbb{R}$ s.t. $f_n' \to g$ uniformly on $I$ as $n \to \infty$.
  Then there exists diffable $f: I \to \mathbb{R}$ s.t. $f_n \to f$ uniformly on $I$ as $n \to \infty$.
  Alternatively $(\lim_{n\to\infty} f_n)' (x) = \lim_{n\to\infty} f_n'(x)$ for all $x \in I$, ie differentiation commutes with taking the uniform limit under our conditions.

### Theorem: Continuity Of Uniform Limits

Let $I \subseteq \mathbb{R}$ be an interval, $f_n, f: I \to \mathbb{R}$ s.t. $f_n \to f$ unifly on $I$ as $n \to \infty$. For all $x_0 \in I$, if each $f_n$ is continuous at $x_0$, then $f$ is continuous at $x_0$.

*Proof.* Let $\epsilon > 0$. Since $f_n \to f$ unifly on $I$, $\exists N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon/3$ for all $x \in I, n \geq N$. Since $f_N$ is cont. at $x_0 \in I$, $\exists \delta > 0$ s.t. $|f_N(x) - f_N(x_0)| < \epsilon/3$ for all $x \in I$ s.t. $|x - x_0| < \delta$. Now let $x \in I$ be s.t. $|x - x_0| < \delta$. Then by the TI, $|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \epsilon$. Thus $f$ is cont. at $x_0 \in I$. □

### Definition: Uniform Convergence Of Series Of Functions

For $E \subseteq \mathbb{R}$, $E \neq \emptyset$, $f_n: E \to \mathbb{R}$ a seq. of funcs; the **sequence of partial sums** is $s_n: E \to \mathbb{R}$, $s_n(x) := \sum_{k=1}^n f_k(x)$ for $x \in E, n \in \mathbb{N}$. $\sum_{n=1}^\infty f_n$
(1) **Converges pointwise on** $E$ if $(s_n)_{n\in\mathbb{N}}$ converges ptwise on $E$.
(2) **Converges uniformly on** $E$ if $(s_n)_{n\in\mathbb{N}}$ converges unifly on $E$
(3) **Cnvges absolutely (ptwise) on** $E$ if $\sum_{n=1}^\infty |f_n(x)|$ cnvges $\forall x \in E$.

### Thm: Continuity of a Uniformly Convergent Series Of Funcs

Let $I \subseteq \mathbb{R}$ be an interval and $f_n, f: I \to \mathbb{R}$ s.t. $\sum_{n=1}^\infty f_n$ converges unfily to $f$ on $I$. For $x_0 \in I$, if each $f_n$ is cont. at $x_0$, $f$ is cont. at $x_0$.

### Theorem: On Termwise Differentiation

Let $I \subset \mathbb{R}$ be a bdd open interval and $f_n: I \to \mathbb{R}$ a seq. of diffable funcs s.t.:
(1) There exists $x_0 \in I$ such that $\sum_{n=1}^\infty f_n(x_0)$ converges
(2) $\sum_{n=1}^\infty f_n'$ converges uniformly on $I$.
Then there exists diffable $f: I \to \mathbb{R}$ such that $\sum_{n=1}^\infty f_n$ converges uniformly to $f$ on $I$, and $f'(x) = \left(\sum_{n=1}^\infty f_n\right)'(x) = \sum_{n=1}^\infty f_n'(x)$ for all $x \in I$.

### Theorem: Weierstrass M-Test

Let $E \subseteq \mathbb{R}$ be $\neq \emptyset$, $f_n: E \to \mathbb{R}$, and $(M_n)_{n\in\mathbb{N}}$ a non-negative seq satisfying:
(1) $|f_n(x)| \leq M_n$ for all $x \in E, n \in \mathbb{N}$
(2) $\sum_{n=1}^\infty M_n$ converges
Then $\sum_{n=1}^\infty f_n$ converges absolutely and uniformly on $E$.

*Proof.* Let $\epsilon > 0$. By the CC for series, $\exists N \in \mathbb{N}$ s.t. $\sum_{k=m+1}^n M_k < \epsilon$ for $n \geq m \geq N$. Thus by hypothesis, $\left|\sum_{k=m+1}^n f_k(x)\right| \leq \sum_{k=m+1}^n |f_k(x)| \leq \sum_{k=m+1}^n M_k < \epsilon$ for all $x \in E, n > m \geq N$. Thus the partial sums of $\sum_{n=1}^\infty f_n$ are unifly Cauchy, so $\sum_{n=1}^\infty f_n$ converges unifly. Moreover, by the above, the partial sums of $\sum_{n=1}^\infty |f_n|$ are unifly Cauchy, so the series converges absolutely. □

## Power Series

### Definition: Power Series

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence and $c \in \mathbb{R}$.
A **power series centred at** $c$ is a series of the form $\sum_{n=0}^\infty a_n(x - c)^n$.

### Definition: Radius Of Convergence $R$

Let $(a_n)_{n\in\mathbb{N}}$ be a real seq and $c \in \mathbb{R}$. Then one of the following holds:
(1) $\exists R > 0, R \in \mathbb{R}$ s.t. $\sum_{n=0}^\infty a_n(x - c)^n$ cnvges absolutely if $|x - c| < R$ and diverges if $|x - c| > R$. Here $R = \sup\{r > 0 : (a_n r^n)_{n\in\mathbb{N}_0}$ is bdd$\}$
(2) $\sum_{n=0}^\infty a_n(x - c)^n$ converges only when $x = c$. Here $R = 0$
(3) $\sum_{n=0}^\infty a_n(x - c)^n$ converges absolutely for all $x \in \mathbb{R}$. Here $R = \infty$
.

### Theorem: On Termwise Differentiation Of Power Series

Let $\sum_{n=0}^\infty a_n(x - c)^n$ have radius of convergence $0 < R \leq \infty$.
Then $f(x) = \sum_{n=0}^\infty a_n(x - c)^n$ for $x \in (c - R, c + R)$ is differentiable on $(c - R, c + R)$; and $f'(x) = \sum_{n=1}^\infty n a_n(x - c)^{n-1}$ for all $x \in (c - R, c + R)$.
- Given the above, $f(x) := \sum_{n=0}^\infty a_n(x - c)^n$ is **infinitely differentiable** on $(c - R, c + R)$, and $a_k = \frac{f^{(k)}(c)}{k!}$ for all $k \in \mathbb{N}_0$.
- For $0 < r < R$, both series converge absolutely and unifly on $[c - r, c + r]$
- For any real sequence $(a_n)_{n\in\mathbb{N}_0}$ and $c \in \mathbb{R}$, $\sum_{n=0}^\infty a_n(x - c)^n$ and $\sum_{n=1}^\infty n a_n(x - c)^{n-1}$ have the same radius of convergence

### Major Theorem: Uniqueness Of Power Series

Power series are unique - For $\sum_{n=0}^\infty a_n(x - c)^n$ and $\sum_{n=0}^\infty b_n(x - c)^n$ with RoCs $0 < R, S, \leq \infty$, if $\exists 0 < r < \min R, S$ s.t. $\sum_{n=0}^\infty a_n(x - c)^n = \sum_{n=0}^\infty b_n(x - c)^n$ for all $x \in \mathbb{R}$ s.t. $|x - c| < r$, then $a_n = b_n$ for all $n \in \mathbb{N}$.

*Proof.* Let $I = (c - r, c + r)$, and $f: I \to \mathbb{R}, f(x) := \sum_{n=0}^\infty a_n(x - c)^n$ for $x \in I$. We know $f$ is infinitely diffable on $I$, and $a_n = \frac{f^{(n)}(c)}{n!}$ for $n \in \mathbb{N}_0$. However, $f(x) = \sum_{n=0}^\infty b_n(x - c)^n$ for $x \in I$, and $b_n = \frac{f^{(n)}(c)}{n!}$ for all $n \in \mathbb{N}_0$. Thus $a_n = b_n$ for all $n \in \mathbb{N}_0$. □

### Definition: Taylor Series

Let $I \subseteq \mathbb{R}$ be a nonempty open interval, and $f: I \to \mathbb{R}$ be infinitely diffable. Then the **Taylor series of $f$ at** $c \in I$ is $\sum_{n=0}^\infty \frac{f^{(n)}(c)}{n!}(x - c)^n$.
$f$ is **real analytic** if its TS converges ptwise on $I$ and equals $f(x)$ at $x$

### Example: Applying The UCT For Integrals

Use the UCT for integrals to prove $\lim_{n\to\infty} \int_0^2 e^{x^2/n}\, dx$ exists and evaluate it.
Let $f_n, f: [o,2] \to \mathbb{R}$ be $f_n(x) := e^{x^2/n}$ and $f(x) := 1$. Clearly both are cont. on $[0,2]$ and thus intable there. Given $0 \leq x \leq 2$, by the MVT $\exists c \in (0,2)$ s.t. $|f_n(x) - f(x)| = \left|e^{x^2/n} - e^0\right| = \frac{|e^c|x^2}{n} \leq \frac{4e^4}{n}$.
Thus $\sup_{x\in[0,2]} |f_n(x) - f(x)| \leq \frac{4c^2}{n} \to 0$ as $n \to \infty$. Then by the squeeze theorem $f_n \to f$ uniformly on $[0,2]$ as $\to \infty$. Since our interval is bounded, by the UCT $\lim_{n\to\infty} \int_0^2 e^{x^2/n}\, dx = \int_0^2 1\, dx = 2$.

### Example: Translation Invariance

Let $E \subseteq \mathbb{R}$ with $m(E) < \infty$, $x \in \mathbb{R}$. Define $x + E = \{x + y : y \in E\}$. Show $m(E) = m(x + E)$.
Let $\epsilon > 0$. By the dfn of measure there exists a seq of open $(I_j)_{j\in\mathbb{N}}$ s.t. $E \subseteq \bigcup_{j=1}^\infty I_j$ and $\sum_{j=1}^\infty \lambda(I_j) < m(E) + \epsilon$. Note the translates $(I_j + x)_{j\in\mathbb{N}}$ are also a seq of open intvls with $\lambda(I_j + x) = \lambda(I_j) \forall j \in \mathbb{N}$. Moreover $E + x \subseteq \bigcup_{j=1}^\infty (I_j + x)$ and $\sum_{j=1}^\infty \lambda(I_j + x) < m(E) + \epsilon$. Taking the infimum we see $m(E + x) \leq m(E) + \epsilon$, since $\epsilon > 0$ arbitrary $m(E + x) \leq m(E)$.
We can apply the same inequality in reverse (since $E = (E + x) - x$ is a translate of $E + x$. Thus $m(E) \leq m(E + x) \leq m(E)$, so $m(E) = m(E + x)$.

### Example: Applying The DCT (Less Evil Example)

Use the DCT to prove $\lim_{n\to\infty} \int_0^\infty (1 + x/n)^{-n} \sin(x/n)\, dx$ exists and evaluate it.
Define $f_n: \mathbb{R} \to \mathbb{R}$ be $f_n(x) := (1 + x/n)^{-n} \sin(x/n) \chi_{[0,\infty)}$, it is the product of a cont. func and the charfunc of an L-mesable set so is L-mesable. $(1 + x/n)^{-n} \to e^x$ ptwise on $\mathbb{R}$ as $n \to \infty$ (proof left as exercise) and $\sin(x/n) \to 0$ ptwise on $\mathbb{R}$ as $n \to \infty$. Thus $f_n(x) \to 0$ ptwise on $\mathbb{R}$ as $n \to \infty$. By observations akin to evil example (binom thm, $n \geq 2$, $(1 + x/n)^n \geq 1 + x^2/4$), $|f_n(x)| \leq g(x) := \frac{4}{4 + x^2}$ for $x \in \mathbb{R}, n \geq 2$. Again by evil example $g$ is L-intable. Thus we can apply the DCT to see that $\lim_{n\to\infty} \int_0^\infty (1 + x/n)^{-n} \sin(x/n)\, dx = \lim_{n\to\infty} \int_{\mathbb{R}} f_n(x)\, dx = \int_{\mathbb{R}} \lim_{n\to\infty} f_n(x)\, dx = \int_0^\infty 0\, dx = 0$.

## Useful Stuff

### Theorem: Random Theorems

- Convergent Geometric Series: For $|r| < 1$, $\frac{1}{1-r} = \sum_{k=0}^\infty r^k$
- $\ln(1 - x) = -\sum_{n=1}^\infty x^n/n$ for $x \in (-1,1)$.
- L'Hôpital's Rule: For interval $c, L \in \mathbb{R} \cup \{\pm\infty\}$, $I \subseteq \mathbb{R}$, $f, g$ diffable on $I\{c\}$ with $g'(x) \neq 0$, and $\lim_{x\to c} f'(x)/g'(x) = L$; if $\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0$ or $\lim_{x\to c} |f(x)| = \lim_{x\to c} |g(x)| = \infty$, $\lim_{x\to c} f(x)/g(x) = \lim_{x\to c} f'(x)/g'(x)$. Also holds for one-sided lims.
- IBP: For $u, v$ cont.sly diffable , $\int uv' = uv - \int u'v$
- Let $f: [a,b] \to \mathbb{R}$ be cont. Then:
  - *MVT*: If $f$ diffable on $(a,b)$; $\exists c \in (a,b)$ s.t. $f'(c)(b - a) = f(b) - f(a)$
  - *IVT*: $\forall x$ between $f(a)$ and $f(b)$, $\exists c \in (a,b)$ s.t. $f(c) = x$.

### Definition: Random Definitions

- $\limsup_{n\to\infty} a_n := \lim_{n\to\infty} (\sup_{k\geq n} a_k)$
- $\liminf_{n\to\infty} a_n := \lim_{n\to\infty} (\inf_{k\geq n} a_k)$
- For $|x - 1| \leq 1$, $x \neq 0$, $\ln(x) = \sum_{k=1}^\infty (-1)^{k-1}(x - 1)^k/k$.

### Remark: Trig Identities

- $\csc\theta = \frac{1}{\sin(\theta)}$, $\sec(\theta) = \frac{1}{\cos(\theta)}$
- $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$
- $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 2\cos^2(\theta) - 1 = 1 - 2\sin^2(\theta)$
- $\sin(x) = \sum_{n=0}^\infty \frac{(-1)^n x^{2n+1}}{(2n+1)!}$
- $\cos(x) = \sum_{n=0}^\infty \frac{(-1)^n x^{2n}}{(2n)!}$.
- $e^x = \sum_{n=0}^\infty \frac{x^n}{n!} = \lim_{n\to\infty} (1 + \frac{x}{n})^n$
- $\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$
- $\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$.
- $\sin'(x) = \cos(x), \cos'(x) = -\sin(x)$
- $\arcsin'(x) = 1/\sqrt{(1 - x^2)}$
- $\arccos'(x) = -1/\sqrt{(1 - x^2)}$
- $\tan'(x) = \sec^2(x)$
- $\arctan'(x) = 1/(1 + x^2)$.
- $\cot'(x) = -\csc^2(x)$
- $\sec'(x) = \sec(x)\tan(x)$
- $\csc'(x) = -\csc(x)\cot(x)$

### Theorem: Convergence Tests

- *Sequences MCT*: If a real sequence $(a_n)_{n\in\mathbb{N}}$ is monotone non-decreasing and bdd above; or monotone non-increasing and bdd below; it converges.
- If a seq converges to $L$, all its subseqs also converge to $L$ (so to disprove convergence find two subseqs converging to different limits).
- *Limsup/inf*: $(x_n)_{n\in\mathbb{N}}$ converges iff $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n$.
- *p-Series*: $\sum_{n=1}^\infty \frac{1}{n^p}$ converges if $p > 1$, diverges otherwise.
- *Ratio*: For $\sum_{n=1}^\infty$ with all $a_n \neq 0$, $\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$; if $0 \leq \rho < 1$ series converges absolutely, if $\rho > 1$ series diverges.
- *Alternating Series*: For $\sum_{n=1}^\infty (-1)^{n+1} b_n$ or $\sum_{n=1}^\infty (-1)^n b_n$, if $b_{n+1} \leq b_n$ for all $n \geq 1$ and $b_n \to 0$ the series converges.
- *Root*: For $\sum_{n+1}^\infty a_n$, $\rho := \lim_{n\to\infty} \sqrt[n]{|a_n|}$, if $0 \leq \rho < 1$ series converges absolutely, if $\rho > 1$ series diverges.
- *Limit Comparison*: For $\sum_{n=1}^\infty a_n$ and $\sum_{n=1}^\infty b_n$, $a_n \geq 0, b_n > 0$ for all $n$, if $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ with $0 < c < \infty$; both series converge or both diverge.

### Example: Applying The Dominated Convergence Theorem

Use the DCT to evaluate $\lim_{n\to\infty} \int_0^1 \frac{1}{(1 + x/n)^n x^{1/n}}\, dx$.
Let $f_n: \mathbb{R} \to \mathbb{R}$ be $f_n(x) := (1 + x/n)^{-n} x^{-1/n} \chi_{(0,\infty)}(x)$ for all $x \in \mathbb{R}$. We claim $f_n$ is L-mesable. Since is is not cont. on $\mathbb{R}$, given $\lambda \in \mathbb{R}$ let $E_\lambda := \{x \in \mathbb{R} : f_n(x) > \lambda\}$. Then if $\lambda < 0$, $E_\lambda = \mathbb{R}$ which is L-mesable. If $\lambda \geq 0$, $E_\lambda = \{x \in (0,\infty) : (1 + x/n)^{-n} x^{-1/n} > \lambda\}$. $g_n: (0,\infty) \to \mathbb{R}$, $g_n(x) := (1 + x/n)^{-n} x^{-1/n}$ for $x > 0$ is a cont. func over an open intvl, so $E_\lambda$ is an open set and thus L-mesable. Thus $f_n(x)$ is L-mesable.
We know $(1 + 1/x)^{-n} \to e^{-x}$ ptwise on $[0,\infty)$, and $x^{-1/n} \to 1$ ptwise on $(0,\infty)$, as $n \to \infty$.
Thus $f(x) := e^{-x} \chi_{(0,\infty)}(x)$ for $x \in \mathbb{R}$ is the ptwise lim of $f_n$ on $\mathbb{R}$.
By the binomial theorem, for $x > 0$, $(1 + \frac{x}{n})^n \geq 1 + n \cdot x/n + \binom{n}{2}\left(\frac{x}{n}\right)^2 = 1 + x + \frac{n(n-1)}{2n^2} \cdot x^2$.
If $n \geq 2$, $n - 1 \geq n/2$, so $(1 + x/n)^n \geq 1 + x^2/4$. It follows from these that $(1 + x/n)^{-n} \leq \frac{4}{4 + x^2}$. Moreover, $x^{1/n} \geq 1$ for $x \geq 1$ and $x^{1/n} \geq x^{1/2}$ for $0 < x \leq 1$ and $n \geq 2$. Combining all these observations: $|f_n(x)| \leq \ell(x) := \frac{4}{4 + x^2} \chi_{[1,\infty)}(x) + \frac{1}{x^{1/2}} \chi_{(0,1)}(x)$ for all $x \in \mathbb{R}, n \geq 2$, and $\ell(x)$ is non-neg and L-mesable.
We claim $\ell$ is L-intable. Indeed by monotonicity of the integral we have $\int_1^\infty \frac{4}{4 + x^2} \chi_{[1,\infty)}(x)\, dx \leq \int_1^\infty \frac{4}{x^2}\, dx$, where the RHS is finite (proof left as an exercise). Moreover define $h: (0,\infty) \to \mathbb{R}$ by $h(x) := x^{-1/2}$ for $x > 0$. Then $H: (0,\infty) \to \mathbb{R}, H(x) := 2x^{1/2}$ for $x > 0$ is a primitive for $h$. By the FTC, for $\epsilon > 0$, $\int_{eps}^1 x^{-1/2}\, dx = \left[2x^{1/2}\right]_\epsilon^1 = 2 - 2\epsilon^{1/2}$.
By the MCT, $\int_{\mathbb{R}} \frac{1}{x^{1/2}} \chi_{(0,1)}(x)\, dx = \int_0^1 \frac{1}{x^{1/2}}\, dx = \lim_{\epsilon\to 0^+} \int_\epsilon^1 \frac{1}{x^{1/2}}\, dx = \lim_{\epsilon\to 0^+} 2 - 2\epsilon^{1/2} = 2$.
Combining these observations with the linearity of the integral, $\int_{\mathbb{R}} \ell(x)\, dx < \infty$, so $\ell$ is L-intable. Thus we may apply the DCT and conclude:
$\lim_{n\to\infty} \int_0^\infty \frac{1}{(1 + x/n)^n x^{1/n}}\, dx = \lim_{n\to\infty} \int_{\mathbb{R}} f_n(x)\, dx = \int_{\mathbb{R}} \lim_{n\to\infty} f_n(x)\, dx = \int_0^\infty e^{-x}\, dx = 1$.